

```
library(ggplot2)
```

Load Data

```
data(mpg)
str(mpg)

## tibble [234 x 11] (S3: tbl_df/tbl/data.frame)
##  $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
##  $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
##  $ displ      : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
##  $ year       : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
##  $ cyl        : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
##  $ trans      : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
##  $ drv       : chr [1:234] "f" "f" "f" "f" ...
##  $ cty       : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
##  $ hwy       : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
##  $ fl        : chr [1:234] "p" "p" "p" "p" ...
##  $ class     : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Convert Numerical Variables to Factors

```
data(mpg)
mpg$cyl <- as.factor(mpg$cyl)
str(mpg)

## tibble [234 x 11] (S3: tbl_df/tbl/data.frame)
##  $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
##  $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
##  $ displ      : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
##  $ year       : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
##  $ cyl        : Factor w/ 4 levels "4","5","6","8": 1 1 1 1 3 3 3 1 1 1 ...
##  $ trans      : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
##  $ drv       : chr [1:234] "f" "f" "f" "f" ...
##  $ cty       : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
##  $ hwy       : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
##  $ fl        : chr [1:234] "p" "p" "p" "p" ...
##  $ class     : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Model

Predict *city mpg* based on *number of cylinders*

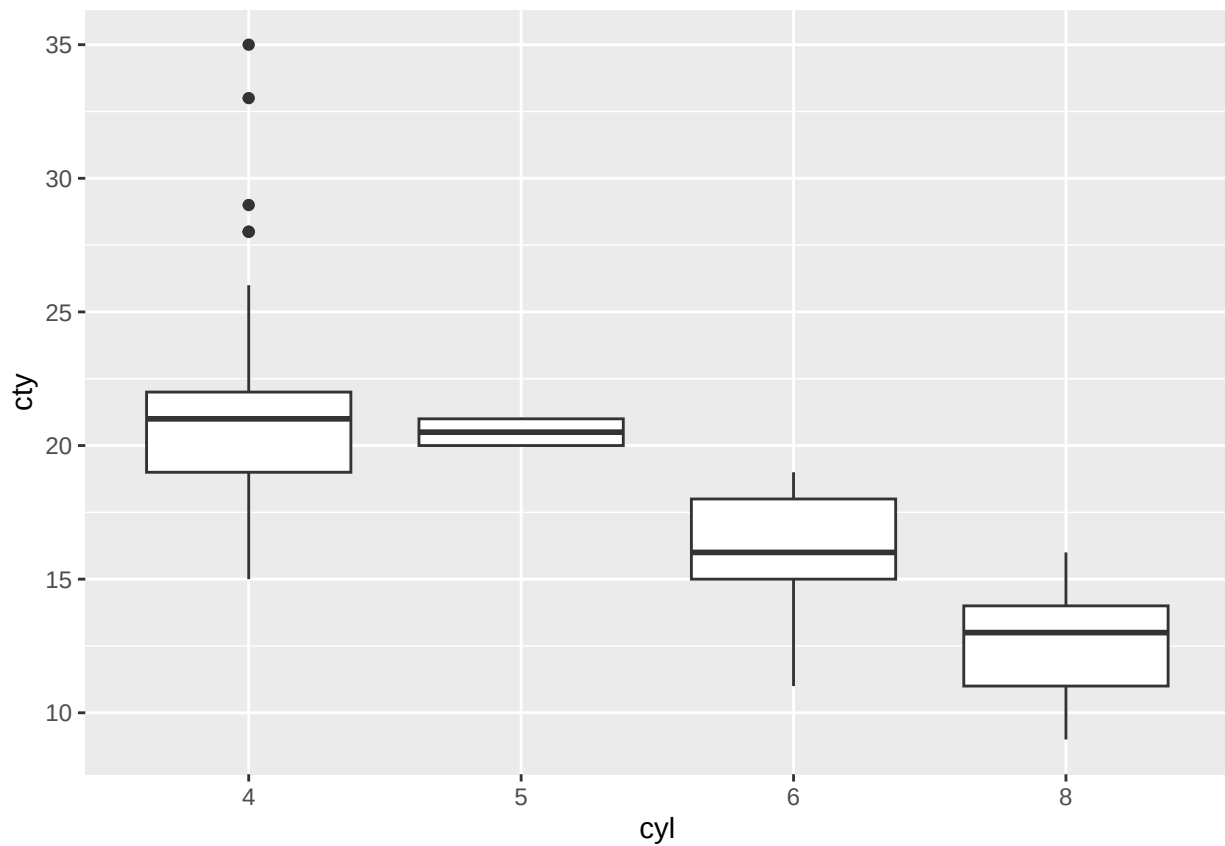
```
model <- lm(cty ~ cyl, data = mpg)
summary(model)

##
## Call:
## lm(formula = cty ~ cyl, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.0123 -1.5714 -0.0123  1.4286 13.9877
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.0123     0.2790  75.305  <2e-16 ***
## cyl5         -0.5123     1.2863  -0.398    0.691
## cyl6         -4.7972     0.3971 -12.081  <2e-16 ***
## cyl8         -8.4409     0.4098 -20.597  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.511 on 230 degrees of freedom
## Multiple R-squared:  0.6563, Adjusted R-squared:  0.6518
## F-statistic: 146.4 on 3 and 230 DF,  p-value: < 2.2e-16
```

Plot

```
ggplot(data = mpg, aes(x = cyl, y = cty)) +
  geom_boxplot()
```



Pairwise Comparison

```
k <- length(unique(mpg$cyl))
k
```

```
## [1] 4
```

```

K <- choose(k, 2)
K

## [1] 6
alpha <- 0.05
astar <- alpha / K
astar

## [1] 0.008333333
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

cyl45 <- filter(mpg, cyl %in% c(4, 5))
# Perform t-test
t45 <- t.test(cty ~ cyl, data = cyl45, conf.level = 1 - astar)
t45

##
## Welch Two Sample t-test
##
## data:   cty by cyl
## t = 1.0582, df = 21.133, p-value = 0.3019
## alternative hypothesis: true difference in means between group 4 and group 5 is not equal to 0
## 99.16667 percent confidence interval:
##  -0.8966594  1.9213508
## sample estimates:
## mean in group 4 mean in group 5
##      21.01235      20.50000

Cannot conclude that there is a difference in mean city mpg between 4 and 5 cylinders.
Fail to reject the null hypothesis.

```

Linear Model Interpretation

hwy miles of 6-cylinder cars

```

model <- lm(hwy ~ cyl, data = mpg)
summary(model)

##
## Call:
## lm(formula = hwy ~ cyl, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.8025 -2.6286  0.1975  2.1975 15.1975

```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  28.80247    0.42880   67.170  <2e-16 ***
## cyl5         -0.05247    1.97666   -0.027    0.979
## cyl6         -5.97968    0.61024   -9.799  <2e-16 ***
## cyl8        -11.17390    0.62978  -17.742  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.859 on 230 degrees of freedom
## Multiple R-squared:  0.5854, Adjusted R-squared:  0.58
## F-statistic: 108.2 on 3 and 230 DF, p-value: < 2.2e-16
intercept <- coefficients(model)["(Intercept)"]
cyl6adjustment <- coefficients(model)["cyl6"]
intercept + cyl6adjustment

## (Intercept)
##      22.82278
```